

Hierarchical Blockchain Architecture for Race Event Management

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Abstract

This document describes a hierarchical blockchain architecture designed for managing race events, such as marathons or running competitions. The system integrates fog computing for low-latency local processing and cloud-based consortium blockchain for scalable, secure data sharing among stakeholders. It ensures data integrity, real-time validation, and auditability while addressing challenges like network latency and privacy concerns.

1 Introduction

In modern race events, accurate tracking of participants, timely validation of checkpoints, and secure sharing of results among organizers, sponsors, and referees are crucial. Traditional systems often suffer from latency issues, data tampering risks, and scalability limitations. To address these, we propose a two-layered blockchain architecture:

- **Private Blockchain Layer (Fog-Enabled):** Operates near the race route for real-time data collection and validation.
- **Consortium Blockchain Network (Cloud-Enabled):** Handles aggregated data for cross-organizational sharing and long-term storage.

The architecture leverages IoT sensors (e.g., RFID for runner bibs), fog nodes at checkpoints, and cloud peers. Data flows from sensors to fog blockchain for local validation, then aggregates to the consortium layer. This ensures low latency at the edge and high trust in the cloud.

Additional considerations include:

- **Security Enhancements:** Use of cryptographic hashing for data immutability and multi-factor authentication for node registration to prevent unauthorized access.
- **Privacy Compliance:** Pseudonymous identities for runners to comply with regulations like GDPR, ensuring personal data is not exposed unnecessarily.
- **Implementation Technologies:** The private layer could use lightweight frameworks like Hyperledger Fabric for fog nodes (e.g., on Raspberry Pi devices), while the consortium layer employs Ethereum-based smart contracts for broader compatibility.
- **Challenges and Mitigations:** Potential issues like intermittent connectivity in remote race areas are mitigated by offline-capable fog nodes that sync when possible. Energy efficiency is achieved through Proof of Stake & Validation (PoSV), reducing computational overhead compared to Proof of Work.

2 Architecture Overview

The system is divided into three logical layers: Edge IoT Layer, Fog-Enabled Private Blockchain, and Cloud-Enabled Consortium Blockchain. Figure 1 illustrates the components and data flow.

Figure 2 shows the deployment in a race scenario, with fog nodes along the route and stakeholders connected via the cloud.

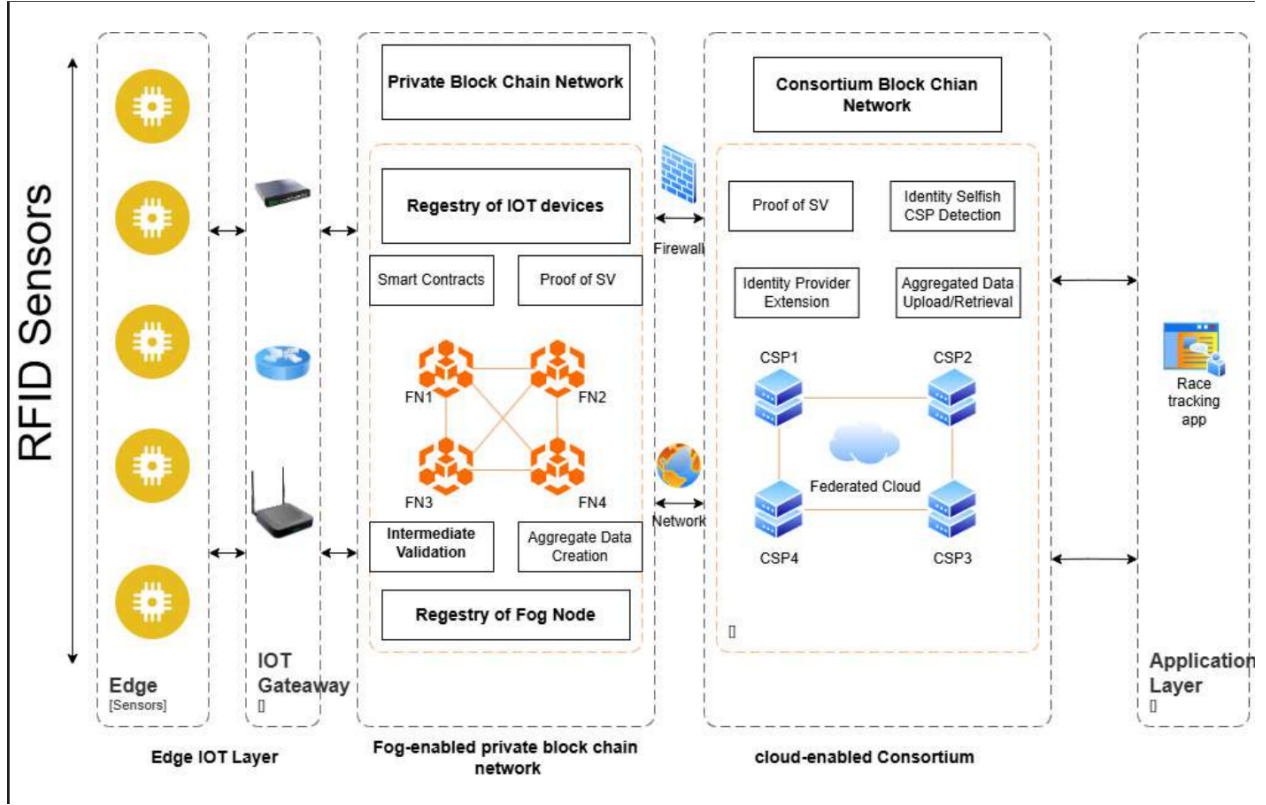


Figure 1: System Architecture Diagram

3 Private Blockchain Layer (Fog-Enabled)

This layer focuses on real-time event recording and validation near the race route, utilizing fog computing for proximity to data sources.

3.1 Purpose

Provides immediate data processing with low latency, ensuring data integrity at the point of collection.

3.2 Key Components

Component	Description and Importance
Fog Nodes / Gateways	Deployed at each checkpoint; collect sensor data and run lightweight blockchain nodes. Reduces latency by processing locally instead of cloud transmission.
Blockchain Structure	Race-specific chain with genesis block at race start; transactions represent checkpoint crossings. Ensures traceability and immutability on-site.
Smart Contracts	Enforce rules like valid runner IDs and sequential checkpoint passage. Automates validation, minimizing human error.

Proof of SV (Stake & Validation)	Hybrid consensus: Nodes vote based on stake and majority agreement. Enables fast, energy-efficient validation suitable for resource-constrained fog environments.
Identity Provider	Assigns secure IDs to runners and sensors linked to bibs. Prevents identity spoofing and ensures authorized data entries.
Aggregate Data Creation	Summarizes multiple readings into batched transactions. Optimizes bandwidth for uploads to the consortium layer.
Registration Module	Onboards fog nodes and bibs pre-race. Establishes a trusted network foundation.
Intermediate Validation	Conducts sanity checks (e.g., timestamp duplicates). Filters invalid data early.

3.3 Example Workflow

When a runner crosses a checkpoint:

1. Sensor reads bib.
2. Fog node validates ID.
3. Smart contract checks sequence.
4. Data added to local block.
5. Nearby nodes reach consensus via PoSV.
6. Batched for consortium upload.

4 Consortium Blockchain Network (Cloud-Enabled)

This layer facilitates collaboration and secure data storage among stakeholders.

4.1 Purpose

Aggregates verified data for sharing, scalability, and auditability across organizations.

4.2 Key Components

Component	Description and Importance
Federated Ledger	Shared blockchain across multiple Cloud Service Providers (CSPs). Distributes trust, preventing single-point control.
Proof of SV	Extended consensus involving CSPs and fog nodes. Maintains integrity during data aggregation.
Identity Provider Extension	Propagates IDs from private to consortium layer. Ensures identity continuity.
Aggregated Data Upload/Retrieval	Handles batched uploads and provides real-time dashboards. Supports scalable querying.
Federated Cloud (CSP Peers)	Each stakeholder hosts a peer node. Enhances resilience and transparency.

Identity Selfish CSP De- tection	Algorithm to identify dishonest CSPs (e.g., data al- teration). Safeguards network reliability.
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4.3 Example Workflow

1. Fog batches upload to consortium.
2. CSPs validate via PoSV.
3. Smart contracts confirm integrity.
4. Stakeholders access verified results.

5 Participants and Their Roles

The system involves various participants, each with defined roles to ensure smooth operation. Table 3 details these.

Participant	Role	Details
Runners	Data Subjects	Wear RFID bibs; their crossings are tracked. Benefit from accurate timing and privacy-protected data.
Sensors (RFID/ID)	Data Collectors	Capture runner data at checkpoints. Ensure precise, tamper-proof readings integrated with IoT gateways.
Fog Nodes	Local Validators	Process and validate data on-site; aggregate for upload. Deployed at checkpoints for low-latency operations.
Organizers	Administrators	Register nodes and bibs; oversee race setup and access dashboards for real-time monitoring.
Referees	Validators	Review aggregated data for disputes; ensure rule compliance via consortium access.
Sponsors	Data Consumers	Access verified results for marketing; host CSP peers for network participation.
Health Team	Emergency Responders	Monitor runner data for health alerts (e.g., anomalies in timings); integrated via CSP.
Betting Company	Analytics Users	Use immutable data for fair betting; connect through consortium for live updates.
Cloud Service Providers (CSPs)	Network Peers	Host blockchain nodes; participate in consensus. Provide scalability and fault tolerance.

Table 3: Participants and Roles

6 How the Two Layers Work Together

The layers interact through a structured process, as outlined in Table 4.

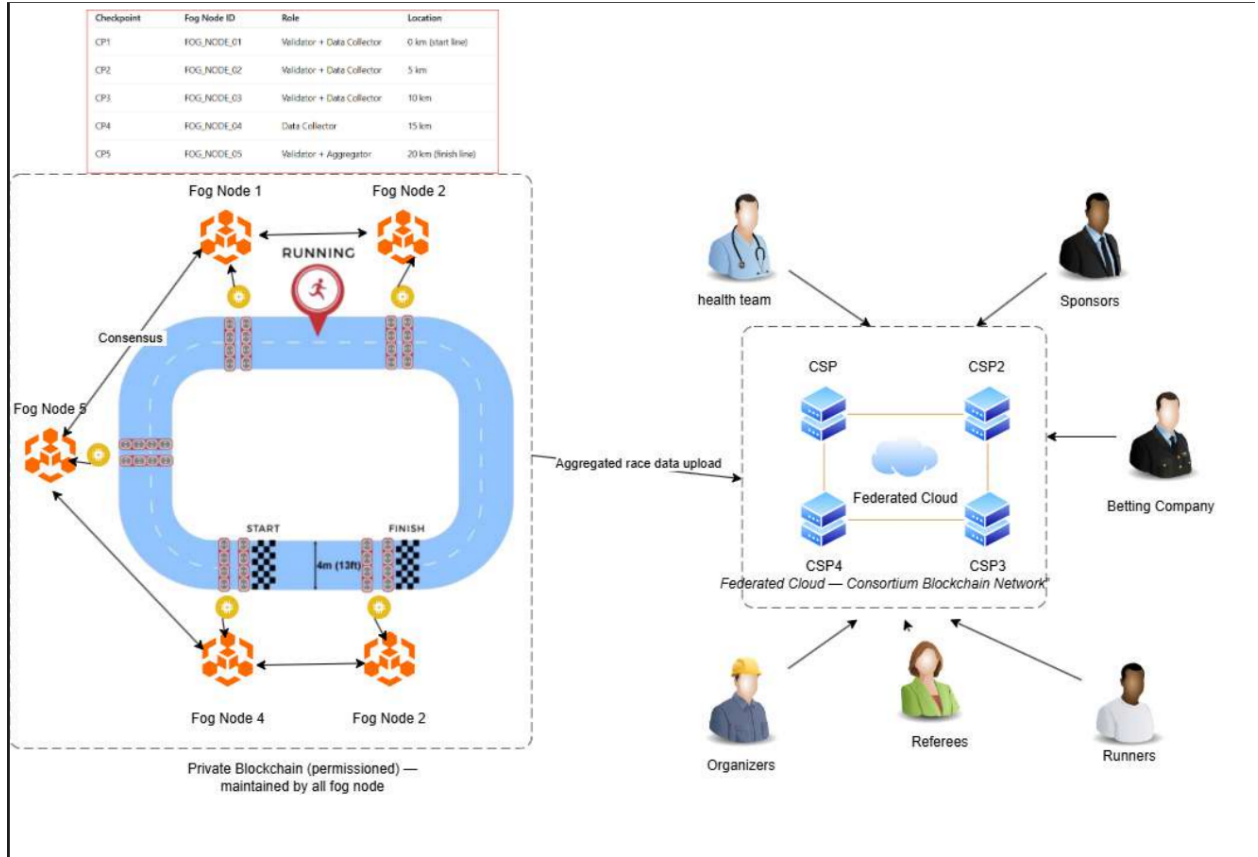


Figure 2: Race Deployment and Stakeholders

7 Benefits of the Architecture

- **Low Latency:** Fog layer enables real-time validation.
- **High Trust:** Immutable audit trails in consortium.
- **Scalability:** Batched data reduces cloud load.
- **Security:** Permissioned access and consensus mechanisms.
- **Fault-Tolerance:** Distributed nodes maintain state.
- **Privacy:** Pseudonymous identities protect users.
- **Cost-Efficiency:** Energy-saving PoSV and edge processing lower operational costs.

In summary, this architecture combines the strengths of fog and cloud computing with blockchain to create a robust system for race management.

Stage	Action	Private Layer (Fog)	Consortium Layer (Cloud)
1. Registration	Setup before race	Registers fog nodes & bibs	Registers organizations & permissions
2. Data Capture	During race	Scans runners locally	Waits for aggregated data
3. Local Validation	Real-time checks	Validates scans instantly via PoSV	—
4. Aggregation	End of stage	Batches verified data	Receives & revalidates
5. Upload	Periodic sync	Sends summaries upward	Updates shared ledger
6. Auditing & Viewing	Post-race	—	Provides dashboards, reports, analytics

Table 4: Interaction Between Layers